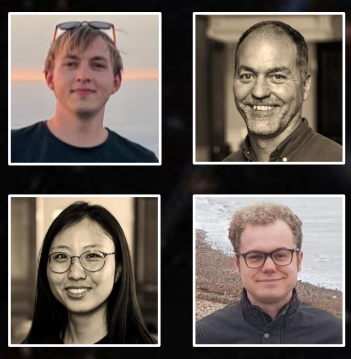




Tracing the dust and ISM properties of distant quiescent galaxies with ALMA and JWST

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DAWN



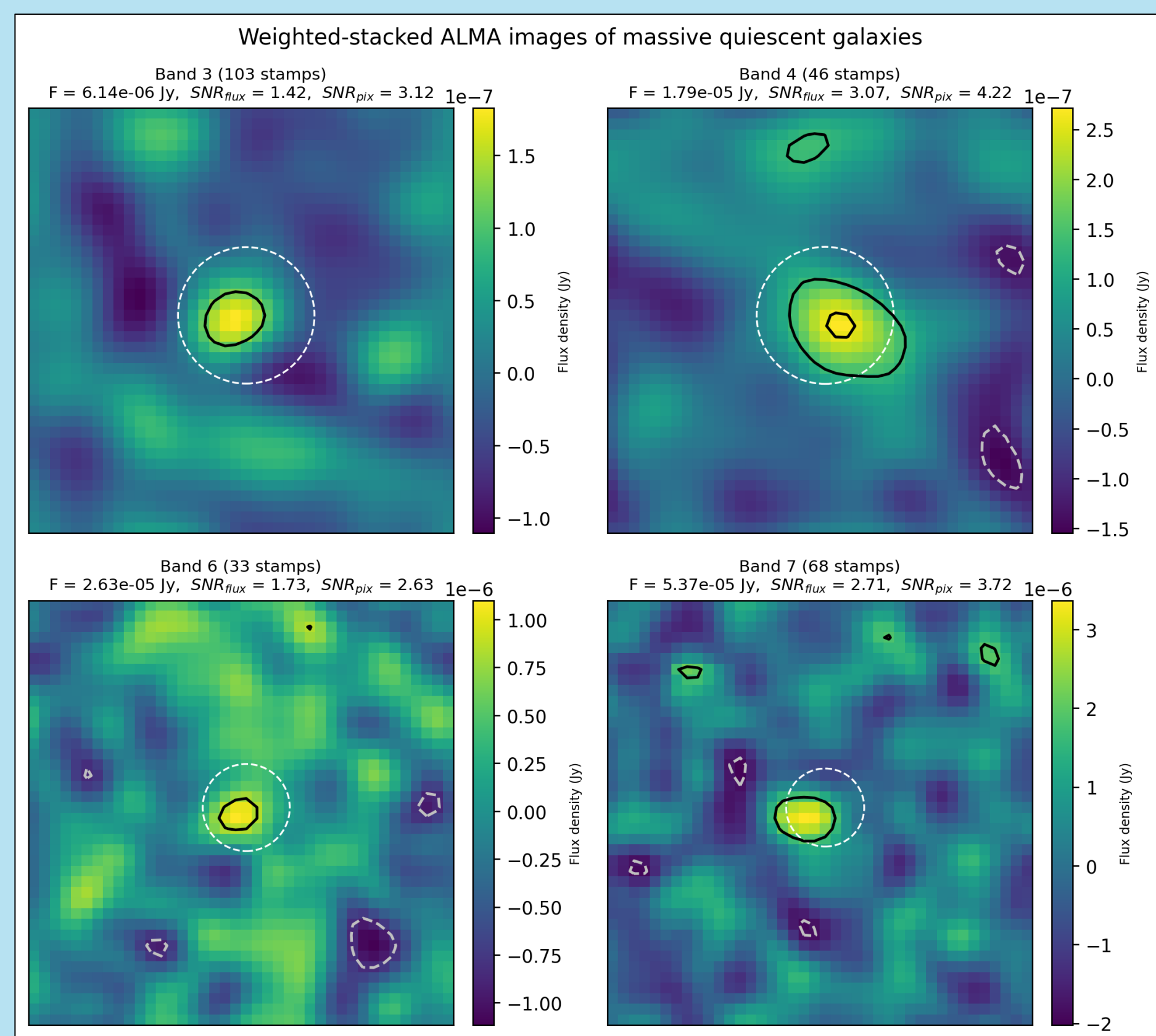
Overview

- Large-scale surveys show that galaxy population is separated into star-forming and quiescent galaxies (QGs)
- Star formation is suppressed by a variety of processes that expel/heat cold gas and dust, siphoning the galaxy of fuel
- JWST has observed an overabundance of QGs in the early universe, challenging current models of cosmology
- I investigate the quenching through two approaches:
 - 1) Constrain how much dust is retained in QGs after quenching through ALMA stacking
 - 2) Characterising AGN and outflow strength using JWST emission line diagnostics

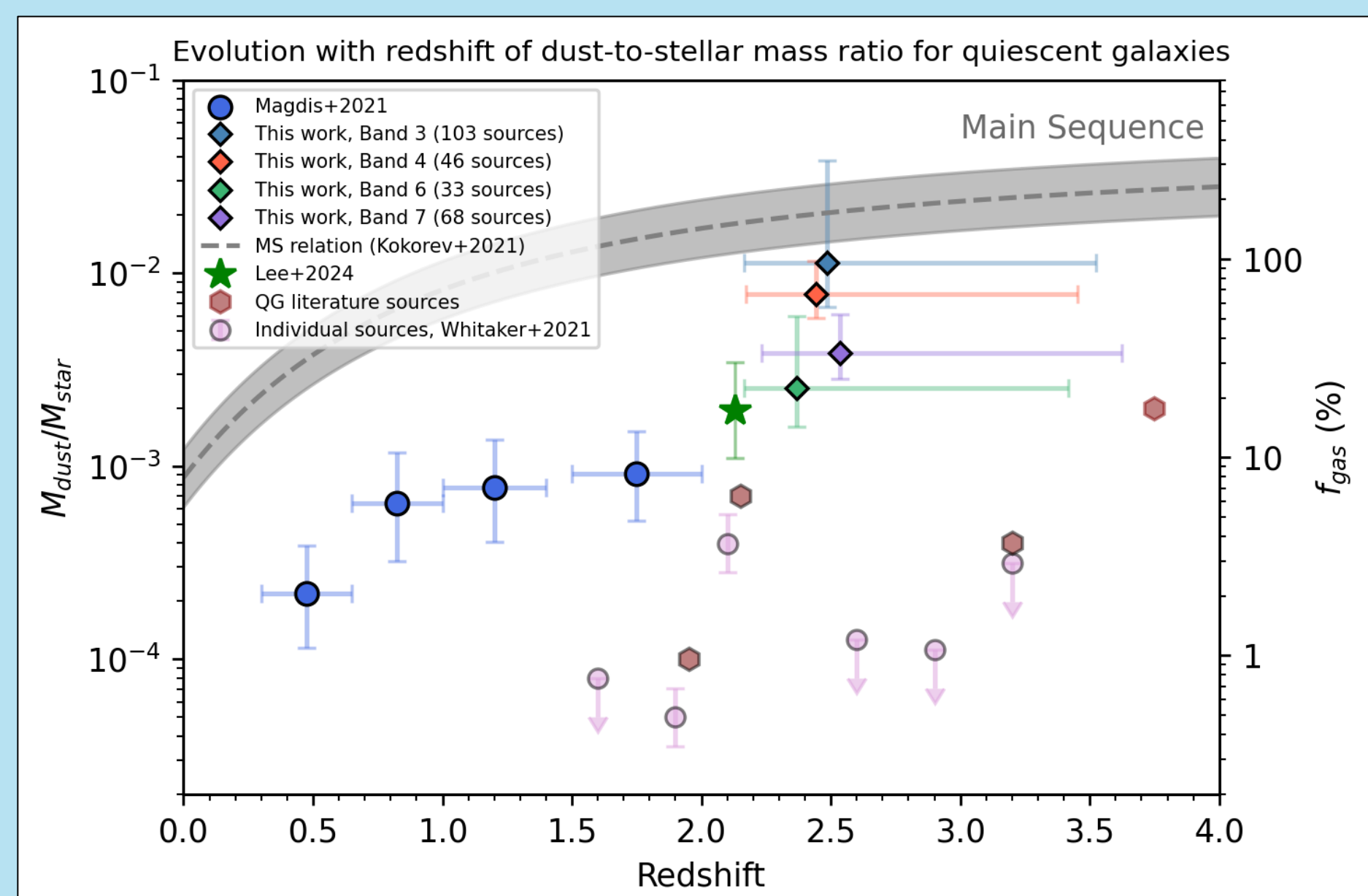


Constraining dust masses with ALMA

- Galaxies are fainter at high-redshift, and individual dust detections are difficult to achieve — However, *stacking* is effective at recovering underlying dust signal
- Sample is 743 galaxies at $z \sim 2-7$ confirmed as quiescent through UVJ colour selection after JWST/NIRCam imaging (Baker et al. 2025b)
- ALMA maps retrieved for 250 sources found by cross-matching source coordinates with archival UDS/COSMOS/GOODS-S maps
- Cutouts are made centered on sources and then stacked, weighted by background noise level (inverse-variance weighting)



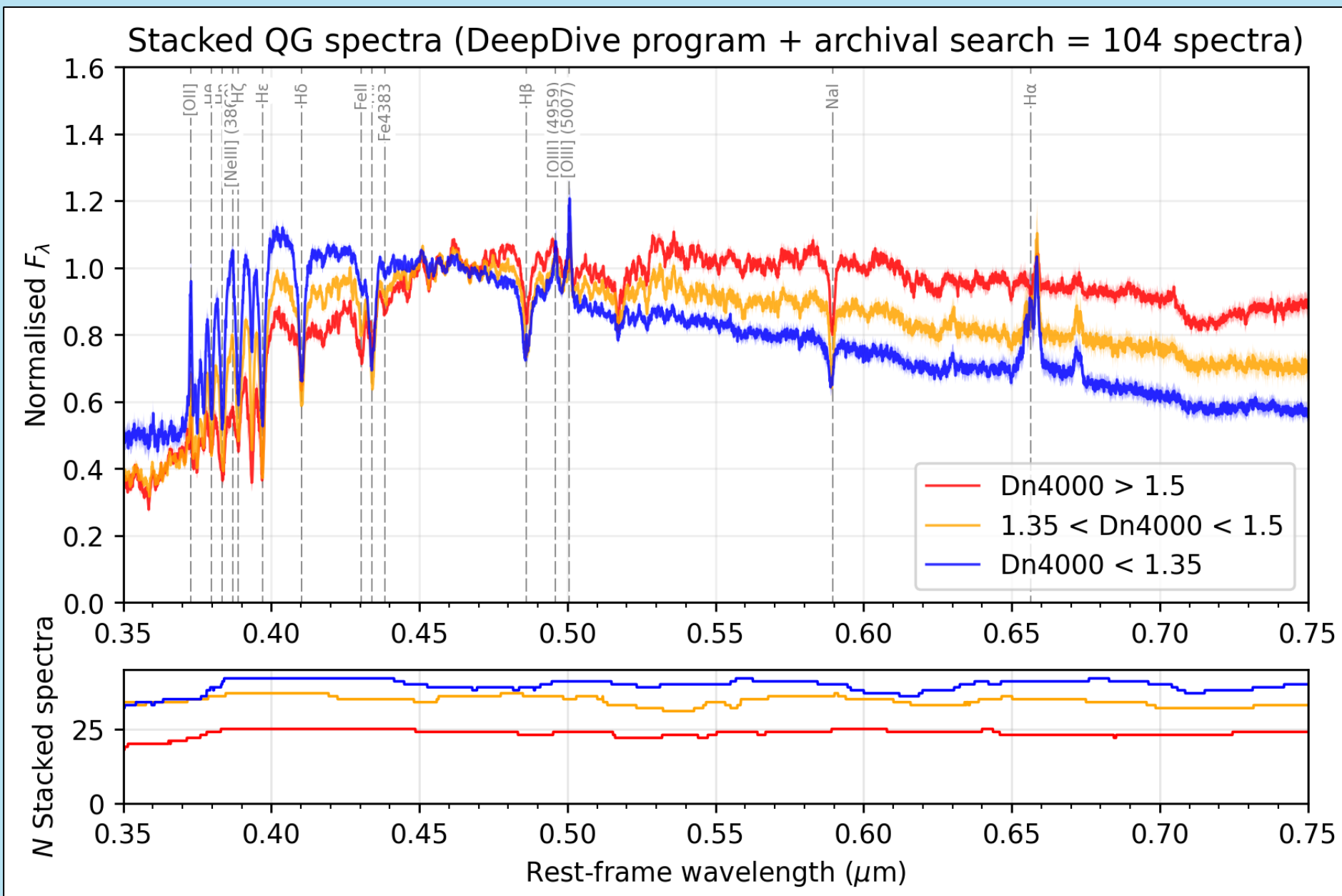
- Fluxes are translated to luminosities, which are compared to a dust SED template for massive QGs to obtain dust masses



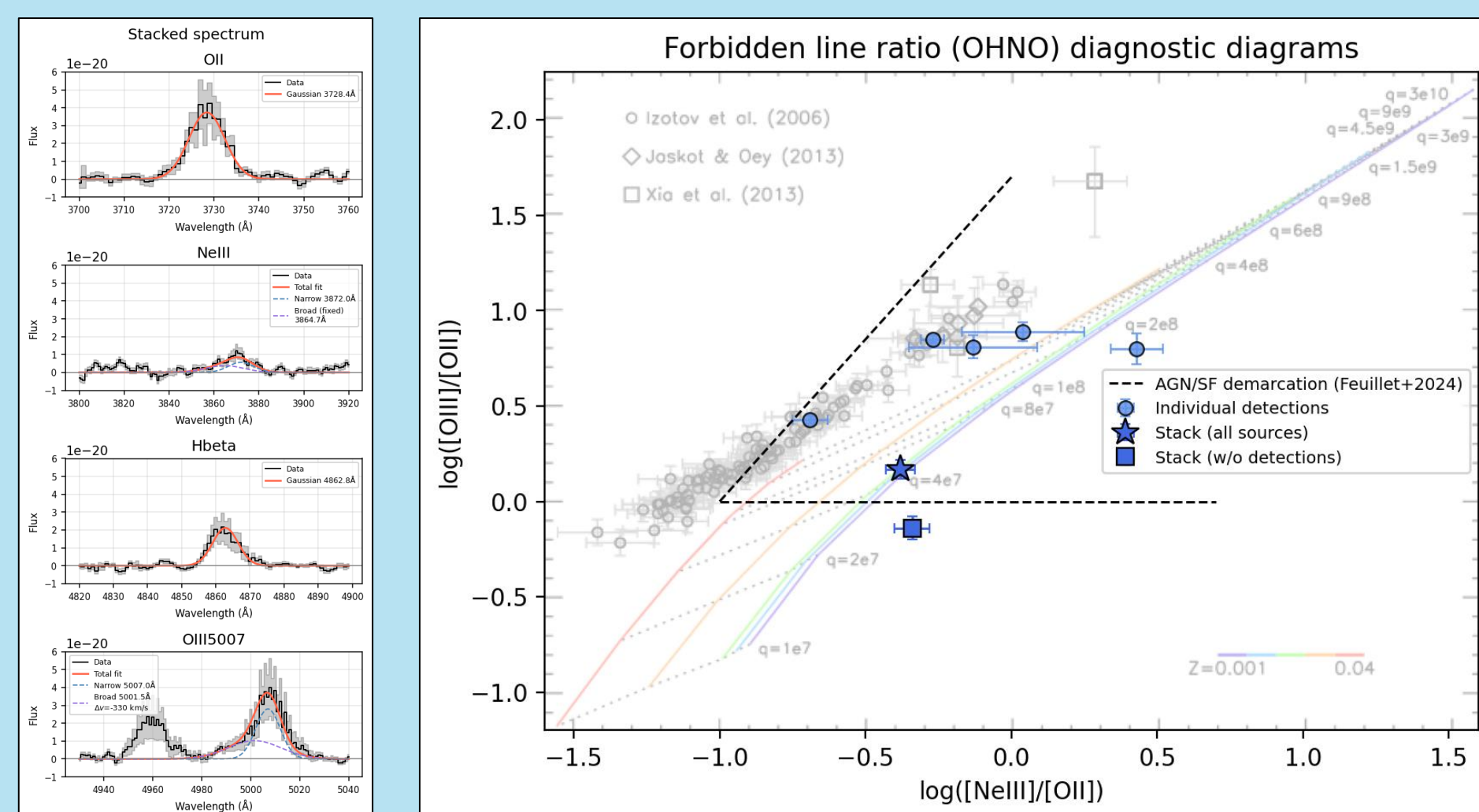
- Interpretation: the dust reservoirs of massive quiescent galaxies might not be depleted as previously thought, although some studies do find high dust-to-stellar-mass ratios in high- z galaxies
- Caveat: results might be viewed as upper limits due to low SNR

Emission line diagnostics with JWST

- Examining the ionisation state of massive QGs is typically done using restframe optical line ratios, however individual detections are challenging at high-redshift
- Similar to measuring dust through ALMA stacking, spectral stacking can recover faint line emissions across the QG population
- 105 JWST/NIRSpec spectra of massive quiescent galaxies at $z \sim 1-4.5$ collected in the *DeepDive* program (Ito et al. 2025)
- Spectra are categorised by their Dn4000 spectral index (proxy for age), re-interpolated to a fixed wavelength scale and stacked, weighted by SNR (favouring less noisy spectra)



- By examining line ratios, we can probe the ionisation conditions of the ISM in QGs and determine the dominant excitation mechanism
- We examine only galaxies with $Dn4000 < 1.35$ as those allow for clearer examination of forbidden lines ([OIII], [NeIII], [OII])



- Strong [NeIII] and [OIII] detections are identified in some individual spectra, which are strong indicators of AGN
- The stack appears to be weighted towards these strong detections, as stacking non-detections shows ISM properties different to that of AGN
- Detection of a broadened blueshifted [OIII] component in the stacked spectrum and in individual spectra, consistent with AGN-driven outflows persisting into the quiescent phase

Conclusion/Future works

- Stacking is a powerful tool to uncover population-level properties of quiescent galaxies that are difficult to robustly detect individually
- Dust reservoirs in QGs may not be depleted as previously thought, with dust-to-stellar-mass ratios suggesting
- JWST emission line diagnostics reveal that ionisation from AGN may be present in some sources, but not over the entire QG population
- Future work: Expanding ALMA coverage to improve dust mass constraints and moving to *uv* stacking for more robust measurements; Applying photoionisation modelling (e.g. CLOUDY, MAPPINGS) to better constrain strain metallicity and ionisation parameter